It seems like you're asking for more details about the tools and commands to configure and manage DNS. Here's a more detailed explanation of the key commands and tools for DNS configuration and management on Windows.

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## 1. `netsh` (Network Shell)

`netsh` is a command-line tool that allows you to configure network interfaces and settings on Windows, including DNS configurations.

### View Current DNS Settings

To view the DNS settings for a specific network interface:

```
netsh interface ipv4 show dns
```

This will list the DNS servers currently assigned to your network interfaces.

### Set DNS Server for an Interface

To set the **primary DNS server** for a specific interface (e.g., Ethernet):

```
netsh interface ipv4 set dns name="Ethernet" static 8.8.8.8
```

- This command sets the DNS server for the `Ethernet` interface to `8.8.8.8`.

### Add Secondary DNS Server

To add a secondary DNS server:

```
netsh interface ipv4 add dns name="Ethernet" 8.8.4.4 index=2
```

- This adds `8.8.4.4` as the **secondary DNS server** for the `Ethernet` interface.

### Remove DNS Server

To remove a DNS server from an interface:

```
netsh interface ipv4 delete dns name="Ethernet" 8.8.8.8
```

- This command removes `8.8.8.8` from the `Ethernet` interface.

### Set DNS to Automatic (Obtain DNS Automatically)

If you want to switch back to automatic DNS settings (using DHCP):

```
netsh interface ipv4 set dns name="Ethernet" source=dhcp
```

---

## 2. PowerShell Cmdlets for DNS Configuration

PowerShell provides more advanced capabilities for managing DNS settings. Below are useful cmdlets for DNS configuration and management.

### View DNS Client Configuration

To view the DNS settings for all network interfaces:

```
Get-DnsClientServerAddress
```

### Set DNS Server for a Specific Interface

To configure DNS servers for a specific interface (use the correct `InterfaceIndex`):

```
Set-DnsClientServerAddress -InterfaceIndex 12 -ServerAddresses ("8.8.8.8",
"8.8.4.4")
```

- Replace `12` with the `InterfaceIndex` of your active network interface. You can find the `InterfaceIndex` by using `Get-NetAdapter`.

### Get Network Interface Index

To get the network interface index:

```
Get-NetAdapter
```

This lists all network adapters and their `InterfaceIndex`, which you can use in the previous command.

### Configure DNS via DHCP (Obtain Automatically)

To set DNS to be obtained automatically via DHCP:

```
Set-DnsClientServerAddress -InterfaceIndex 12 -ResetServerAddresses
```

### Clear DNS Cache

To clear the DNS cache:

```
Clear-DnsClientCache
```

---

## 3. `nslookup` (Name Server Lookup)

`nslookup` is a diagnostic tool for querying DNS servers to obtain information about domain names, IP addresses, and DNS records.

**Basic Query**

To query a DNS server for the IP address of a domain:

```
nslookup example.com
```

This will query your default DNS server for the IP address of `example.com`.

**Query with a Specific DNS Server**

To query a specific DNS server (e.g., Google's DNS server `8.8.8.8`):

```
nslookup example.com 8.8.8.8
```

**Interactive Mode**

To use `nslookup` in interactive mode:

```
nslookup
```

In interactive mode, you can type multiple queries and switch DNS servers. For example:

```
set type=MX
example.com
```

This command queries the MX (Mail Exchange) records for `example.com`.

**Query for Specific DNS Record Types**

To query for specific DNS record types, you can use the `set type` command:

```
nslookup
set type=MX
example.com
```

This queries the mail exchange (MX) records for `example.com`.

---

## 4. `ipconfig` (Internet Protocol Configuration)

`ipconfig` is a basic network configuration tool. It’s helpful for managing IP addresses, releasing/renewing DHCP leases, and viewing DNS settings.

### View DNS Settings

To view your current DNS settings, use:

```
ipconfig /all
```

This command shows detailed information about all network interfaces, including DNS servers assigned to each.

### Flush DNS Cache

If you're experiencing issues with DNS resolution, flushing the cache might help:

```
ipconfig /flushdns
```

This clears the local DNS resolver cache, which stores previously resolved domain names and IP addresses.

### Display DNS Cache

To view the contents of the DNS cache:

```
ipconfig /displaydns
```

This will show the cached DNS entries on your system.

### Release and Renew IP Address

To release your current IP address (if using DHCP) and renew it:

```
ipconfig /release
ipconfig /renew
```

This can also refresh the DNS configuration if there are any issues.

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## 5. DNS Server Management Tools (Windows Server)

If you're running a **DNS server** on a Windows Server machine, you can manage DNS zones and records using either the **DNS Manager** graphical interface or the **`dnscmd`** command-line tool.

### `dnscmd` Command-line Tool

The `dnscmd` tool is used to manage DNS servers, zones, and records on Windows Server.

#### View All Zones

To list all DNS zones on your DNS server:

```
dnscmd /enumzones
```

#### Create a New DNS Zone

To create a new **primary DNS zone** for a domain:

```
dnscmd /zoneadd example.com /primary
```

#### Add a DNS Record

To add an **A record** (address record) for a domain:

```
dnscmd /recordadd example.com @ A 192.0.2.1
```

This adds an **A record** for `example.com`, pointing to the IP address `192.0.2.1`.

#### Delete a DNS Record

To delete an existing record:

```
dnscmd /recorddelete example.com @ A 192.0.2.1
```

#### Add an MX Record (Mail Exchange)

To add an **MX record** for mail routing:

```
dnscmd /recordadd example.com @ MX 10 mail.example.com
```

This adds an **MX record** for `example.com`, with a mail exchange priority of `10` pointing to `mail.example.com`.

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## 6. Troubleshooting DNS

When troubleshooting DNS issues, you can use several tools to identify and resolve problems.

### Using `nslookup` for Troubleshooting

If you're having trouble resolving a domain, `nslookup` can help diagnose the issue by querying a specific DNS server.

#### Check if DNS is Resolving

```
nslookup example.com
```

This checks if `example.com` can be resolved by your current DNS server.

### Using `ping` to Check Network Connectivity

If DNS resolution works but you're still unable to connect, try using `ping` to check if the target host is reachable:

```
ping example.com
```

### Check DNS Server Availability

To check if a DNS server is responding:

```
ping 8.8.8.8
```

This pings Google's DNS server to ensure that it’s reachable.

### Using `tracert` for Path Diagnosis

If you're having network issues, `tracert` can help identify where the connection is failing along the path:

```
tracert example.com
```

---

## Summary of Key DNS Commands

| Tool/Command | Description |
|---|---|
| `netsh interface ipv4 set dns` | Set DNS servers for a specific network interface. |
| `Set-DnsClientServerAddress` | PowerShell cmdlet to configure DNS server addresses. |
| `nslookup` | Query DNS servers to troubleshoot or resolve domain names. |
| `ipconfig /all` | View the DNS settings and network configuration. |
| `ipconfig /flushdns` | Clear the DNS cache. |
| `dnscmd` | Manage DNS zones and records (Windows Server). |
| `ping` | Test network and DNS resolution. |
| `tracert` | Trace the route taken to reach a target host. |